

Week 4

Everything “is” matrix multiplication:

matrix multiplication	covector / vector / scalar version		
Example	Example	operation	
$\underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{2 \times 1} = \underbrace{\begin{bmatrix} 1 \\ 5 \\ 9 \end{bmatrix}}_{3 \times 1}$	$\underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\in \mathbb{R}^2} = \underbrace{\begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix}}_{\in \mathbb{R}^3}$	$\underbrace{Ax}_{\in \mathbb{R}^m}$	matrix-vector multiplication $A \in \mathbb{R}^{m \times n}, \mathbf{x} \in \mathbb{R}^n$ (Definition 2.4)
$\underbrace{\begin{bmatrix} 1 & 2 \end{bmatrix}}_{1 \times 2} \underbrace{\begin{bmatrix} 3 \\ 4 \end{bmatrix}}_{2 \times 1} = \underbrace{\begin{bmatrix} 11 \end{bmatrix}}_{1 \times 1}$	$\underbrace{\begin{pmatrix} 1 & 2 \end{pmatrix}}_{\in (\mathbb{R}^2)^*} \underbrace{\begin{pmatrix} 3 \\ 4 \end{pmatrix}}_{\in \mathbb{R}^2} = \underbrace{11}_{\in \mathbb{R}}$	$\underbrace{\mathbf{v}^\top \mathbf{w}}_{\in \mathbb{R}}$	scalar product $\mathbf{v}, \mathbf{w} \in \mathbb{R}^m$ (Definition 1.9, Definition 1.19)
$\underbrace{\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}}_{1 \times 3} \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} = \underbrace{\begin{bmatrix} 6 & 9 \end{bmatrix}}_{1 \times 2}$	$\underbrace{\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}}_{\in (\mathbb{R}^3)^*} \underbrace{\begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix}}_{3 \times 2} = \underbrace{\begin{pmatrix} 6 & 9 \end{pmatrix}}_{\in (\mathbb{R}^2)^*}$	$\underbrace{\mathbf{y}^\top A}_{\in (\mathbb{R}^n)^*}$	covector-matrix multiplication $\mathbf{y}^\top \in (\mathbb{R}^m)^*, A \in \mathbb{R}^{m \times n}$ (Definition 2.43)
$\underbrace{\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}}_{3 \times 1} \underbrace{\begin{bmatrix} 1 & 2 \end{bmatrix}}_{1 \times 2} = \underbrace{\begin{bmatrix} 3 & 6 \\ 4 & 8 \\ 5 & 10 \end{bmatrix}}_{3 \times 2}$	$\underbrace{\begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}}_{\in \mathbb{R}^3} \underbrace{\begin{pmatrix} 1 & 2 \end{pmatrix}}_{\in (\mathbb{R}^2)^*} = \underbrace{\begin{bmatrix} 3 & 6 \\ 4 & 8 \\ 5 & 10 \end{bmatrix}}_{3 \times 2}$	$\underbrace{\mathbf{v} \mathbf{w}^\top}_{m \times n}$	outer product $\mathbf{v} \in \mathbb{R}^m, \mathbf{w} \in \mathbb{R}^n$ (Definition 2.44)
$\underbrace{\begin{bmatrix} 3 \\ 4 \end{bmatrix}}_{2 \times 1} \underbrace{\begin{bmatrix} 2 \end{bmatrix}}_{1 \times 1} = \underbrace{\begin{bmatrix} 6 \\ 8 \end{bmatrix}}_{2 \times 1}$	$\underbrace{\begin{pmatrix} 3 \\ 4 \end{pmatrix}}_{\in \mathbb{R}^2} \underbrace{2}_{\in \mathbb{R}} = \underbrace{\begin{pmatrix} 6 \\ 8 \end{pmatrix}}_{\in \mathbb{R}^2}$	$\underbrace{\mathbf{v} \lambda}_{\in \mathbb{R}^m}$	scalar multiplication (the other way around) $\mathbf{v} \in \mathbb{R}^m, \lambda \in \mathbb{R}$ (Definition 1.3).
$\underbrace{\begin{bmatrix} 2 \end{bmatrix}}_{1 \times 1} \underbrace{\begin{bmatrix} 3 & 4 \end{bmatrix}}_{1 \times 2} = \underbrace{\begin{bmatrix} 6 & 8 \end{bmatrix}}_{1 \times 2}$	$\underbrace{2}_{\in \mathbb{R}} \underbrace{\begin{pmatrix} 3 & 4 \end{pmatrix}}_{\in (\mathbb{R}^2)^*} = \underbrace{\begin{pmatrix} 6 & 8 \end{pmatrix}}_{\in (\mathbb{R}^2)^*}$	$\underbrace{\lambda \mathbf{v}^\top}_{\in (\mathbb{R}^m)^*}$	scalar multiplication (covector version) $\mathbf{v}^\top \in \mathbb{R}^m, \lambda \in \mathbb{R}$
$\underbrace{\begin{bmatrix} 2 \end{bmatrix}}_{1 \times 1} \underbrace{\begin{bmatrix} 3 \end{bmatrix}}_{1 \times 1} = \underbrace{\begin{bmatrix} 6 \end{bmatrix}}_{1 \times 1}$	$\underbrace{2}_{\in \mathbb{R}} \underbrace{3}_{\in \mathbb{R}} = \underbrace{6}_{\in \mathbb{R}}$	$\underbrace{\lambda \mu}_{\in \mathbb{R}}$	standard multiplication $\lambda, \mu \in \mathbb{R}$

(Mixed) products (e.g. $\mathbf{x}^\top A^\top A \mathbf{x}$) can be evaluated (using matrix multiplication) as if

- vectors in $\mathbb{R}^m$ were $m \times 1$ matrices,
- covectors in $(\mathbb{R}^n)^*$ were $1 \times n$ matrices,
- scalars in $\mathbb{R}$ were 1×1 matrices.

Only $[\dots]$ must suitably be adapted to $(\dots)$ or $\dots$.

Associativity still holds, bracketing does not matter (Lemma 2.42).

Transpose a product: transpose individual factors and multiply them in reverse order (Lemma 2.40); $\lambda^\top = \lambda$.

Examples:

Alternative bracketings:

$$\underbrace{\underbrace{\mathbf{x}^\top A}_{\text{covector vector}} \underbrace{A \mathbf{x}}_{\text{vector}}}_{\text{scalar}} = (A \mathbf{x})^\top (A \mathbf{x}) = \|A \mathbf{x}\|^2$$

$$\underbrace{\underbrace{\mathbf{x}^\top}_{\text{covector}} \underbrace{A^\top A}_{\text{matrix}} \underbrace{\mathbf{x}}_{\text{vector}}}_{\text{scalar}} \quad \text{or} \quad \underbrace{\underbrace{\mathbf{x}^\top}_{\text{covector}} \underbrace{A^\top A}_{\text{matrix}} \underbrace{\mathbf{x}}_{\text{vector}}}_{\text{scalar}}$$

$$\underbrace{\underbrace{\mathbf{v}^\top \mathbf{w}}_{\text{scalar}} \underbrace{\mathbf{v}^\top \mathbf{w}}_{\text{scalar}}}_{\text{scalar}} = (\mathbf{v}^\top \mathbf{w})^2$$

$$\underbrace{\underbrace{\mathbf{v}^\top}_{\text{covector}} \underbrace{\mathbf{w} \mathbf{v}^\top}_{\text{matrix}} \underbrace{\mathbf{w}}_{\text{vector}}}_{\text{scalar}} \quad \leftarrow \begin{array}{l} \uparrow \\ \text{may be very} \\ \text{inefficient!} \end{array}$$

Most efficient bracketing can be computed by *dynamic programming*.

Matrix multiplication...

...in column notation (Definition 2.36)

...in row notation (Observation 2.45).

$$A \underbrace{\begin{bmatrix} | & | & \cdots & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_b \\ | & | & \cdots & | \end{bmatrix}}_B = \underbrace{\begin{bmatrix} | & | & \cdots & | \\ A \mathbf{x}_1 & A \mathbf{x}_2 & \cdots & A \mathbf{x}_b \\ | & | & \cdots & | \end{bmatrix}}_{AB}$$

$$\underbrace{\begin{bmatrix} - & \mathbf{u}_1^\top & - \\ - & \mathbf{u}_2^\top & - \\ & \vdots & \\ - & \mathbf{u}_a^\top & - \end{bmatrix}}_A B = \underbrace{\begin{bmatrix} - & \mathbf{u}_1^\top B & - \\ - & \mathbf{u}_2^\top B & - \\ & \vdots & \\ - & \mathbf{u}_a^\top B & - \end{bmatrix}}_{AB}$$

CR decomposition:

Prime number decomposition: $1001 = 7 \cdot 11 \cdot 13$

Matrix decomposition: write a matrix A as product of matrices to understand A better!

Theorem 2.46: Let A be an $m \times n$ matrix of rank r (Definition 2.10). Let C be the $m \times r$ submatrix of A containing the independent columns. Then there is a unique $r \times n$ matrix R' such that

$$A = CR'.$$

Example ($r = 1$):

$$\underbrace{\begin{bmatrix} 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 2 \\ 3 \end{bmatrix}}_{C, 2 \times 1} \underbrace{\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}}_{R', 1 \times 3}.$$

Columns of R' : scalars to write each column of A as a scalar multiple of the first column.

Proof. A and C have the same column space (Lemma 2.11) $\Rightarrow$ every column of A is a linear combination of the columns of C :

$$\begin{array}{ccc} \mathbf{v}_j = C \mathbf{x}_j & : & \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{bmatrix} = C \underbrace{\begin{bmatrix} | & | & \cdots & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_n \\ | & | & \cdots & | \end{bmatrix}}_{R' \in \mathbb{R}^{r \times n}} \end{array}$$

column j of A vector of r scalars A $R' \in \mathbb{R}^{r \times n}$

The columns of C are linearly independent: all $\mathbf{x}_j$ are unique (Lemma 1.24). So R' is unique. $\square$

Example:

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \quad \begin{array}{c} \text{columns of } A \\ \mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{v}_4 \\ \parallel \quad \parallel \quad \parallel \quad \parallel \\ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \\ \parallel \quad \parallel \quad \parallel \quad \parallel \\ 1\mathbf{v}_1 \quad 2\mathbf{v}_1 \quad \quad \quad 3\mathbf{v}_1 \\ \quad \quad \quad \quad \quad 1\mathbf{v}_3 \quad -2\mathbf{v}_3 \\ \text{independent?} \quad \text{yes} \quad \text{no} \quad \text{yes} \quad \text{no} \end{array}$$

$$\underbrace{\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix}}_{A, 3 \times 4} = \underbrace{\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}}_{C, 3 \times 2} \underbrace{\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}}_{R', 2 \times 4}$$

Here we proved existence of $A = CR'$, *not* efficient computation (see Chapter 3).

$A = CR'$ and matrix compression:

$m \times n$ matrix A : mn entries

$m \times n$ matrix A of rank r : C has mr entries, R' has rn entries: $(m+n)r$ entries

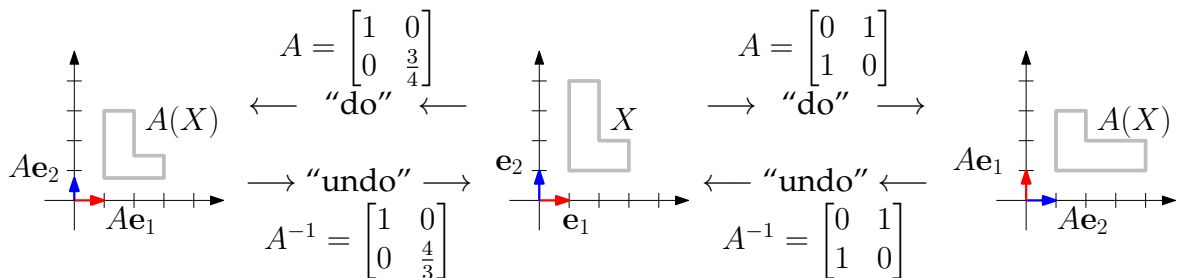
If r is small, storing / transmitting C, R' instead of A is more efficient.

$$m = n = 1,000, r = 10 : \quad \begin{array}{ll} mn : & 1,000,000 \\ (m+n)r : & 20,000 \end{array}$$

Invertible and inverse matrices (Section 2.4)

$T_A : \text{input} \mapsto \text{output}$ ("do").

$? : \text{output} \mapsto \text{input}$ ("undo")



$A = 0$: $T_A : \text{input} \mapsto 0$, cannot "undo" (no idea what the input was).

Definition 2.48: Let X, Y be sets and $f : X \rightarrow Y$ a function.

- (i) f is *injective* if for every $y \in Y$, there is at most one $x \in X$ with $f(x) = y$.
("For every possible output, at most one input leads to it.")
- (ii) f is *surjective* if for every $y \in Y$, there is at least one $x \in X$ with $f(x) = y$.
("For every possible output, at least one input leads to it.")
- (iii) f is *bijective* (undoable) if f is both injective and surjective.
("For every possible output, exactly one input leads to it.")
- (iv) The *inverse* of a bijective function f is the function

$$f^{-1} : Y \rightarrow X, \quad y \mapsto \text{the unique } x \in X \text{ such that } f(x) = y.$$

Fact 2.49: If $f : X \rightarrow Y$ is bijective, then $f^{-1} : Y \rightarrow X$ is also bijective, and $(f^{-1})^{-1} = f$. Moreover, $f^{-1} \circ f = \text{id}$ (f^{-1} is undoing f) and $f \circ f^{-1} = \text{id}$ (f is undoing f^{-1}).

$$\text{id} : \text{input} \mapsto \text{input}$$

Matrix transformation $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \mathbf{x} \mapsto A\mathbf{x}$:

- $m < n$ (A wide): not injective (Lemma 2.50): "More inputs than possible outputs."
- $m > n$ (A tall): not surjective (Lemma 2.51): "More possible outputs than inputs."
- $m = n$ (A square): potentially bijective, but not always (see $A = 0 \in \mathbb{R}^{m \times m}$)

Lemma 2.52: Let $T : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a bijective linear transformation. Then its inverse $T^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is also a linear transformation (and bijective by Fact 2.49).

Proof. Prove linearity of T^{-1} (Definition 2.21):

$$T^{-1}(\lambda_1 \mathbf{y}_1 + \lambda_2 \mathbf{y}_2) = \lambda_1 T^{-1}(\mathbf{y}_1) + \lambda_2 T^{-1}(\mathbf{y}_2).$$

$$\mathbf{x}_1 := T^{-1}(\mathbf{y}_1), \mathbf{x}_2 := T^{-1}(\mathbf{y}_2).$$

$$(T(\mathbf{x}_1) = \mathbf{y}_1 \text{ and } T(\mathbf{x}_2) = \mathbf{y}_2.)$$

Linearity of T :

$$T(\lambda_1 \underbrace{T^{-1}(\mathbf{y}_1)}_{\mathbf{x}_1} + \lambda_2 \underbrace{T^{-1}(\mathbf{y}_2)}_{\mathbf{x}_2}) = \lambda_1 \underbrace{\mathbf{y}_1}_{T(\mathbf{x}_1)} + \lambda_2 \underbrace{\mathbf{y}_2}_{T(\mathbf{x}_2)}.$$

Apply T^{-1} to both sides (and use $T^{-1}(T(\text{input})) = \text{input}$):

$$\lambda_1 T^{-1}(\mathbf{y}_1) + \lambda_2 T^{-1}(\mathbf{y}_2) = T^{-1}(\lambda_1 \mathbf{y}_1 + \lambda_2 \mathbf{y}_2).$$

□

How can we tell from A whether T_A is bijective?

Lemma 2.53: Let A be an $m \times m$ matrix. The following three statements are equivalent.

- (i) $T_A : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is bijective.
- (ii) There is an $m \times m$ matrix B such that $BA = I$.
- (iii) The columns of A are linearly independent.

Proof structure: (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i):

Proof.

(i) $\Rightarrow$ (ii): If T_A is bijective, its inverse T_A^{-1} (Definition 2.48) is a matrix transformation T_B : T_A is a linear transformation (Observation 2.22), so T_A^{-1} is also linear transformation (Lemma 2.52) and a matrix transformation (Theorem 2.26). Fact 2.49:

$$\underbrace{T_B \circ T_A}_{\text{Cor. 2.37 } T_{BA}} = \underbrace{\text{id}}_{T_I}.$$

By Theorem 2.26 (unique matrix): $BA = I$.

(ii) $\Rightarrow$ (iii): If $BA = I$, then $\mathbf{x} = \mathbf{0}$ is the only vector with $A\mathbf{x} = \mathbf{0}$ (so the columns of A are linearly independent by Obs. 2.5): Let $A\mathbf{x} = \mathbf{0}$. Then $\mathbf{x} = (BA)\mathbf{x} = B(A\mathbf{x}) = B\mathbf{0} = \mathbf{0}$.

(iii) $\Rightarrow$ (i): If the columns of A are linearly independent, T_A is surjective: for each $\mathbf{y} \in \mathbb{R}^m$, there is at least one $\mathbf{x} \in \mathbb{R}^m$ with $T_A(\mathbf{x}) = A\mathbf{x} = \mathbf{y}$. True because the m columns of A span $\mathbb{R}^m$ (Lemma 1.28).

T_A is even bijective: there is exactly one such $\mathbf{x}$. True because $\mathbf{y}$ can be written as a linear combination of the columns of A in only one way (Lemma 1.24). $\square$

In (ii), we could as well have said " $AB = I$ ":

Lemma 2.54: Let A, B be $m \times m$ matrices such that $BA = I$. Then also $AB = I$.

Definition and basic properties:

Definition 2.55: Let A be an $m \times m$ matrix. A is called *invertible* if there is an $m \times m$ matrix B such that $BA = I$. (By Lemma 2.53 this is the case if and only if A has linearly independent columns.) Otherwise, A is called *singular*.

Observation 2.56: In the definition, we could as well have said " $AB = I$ " or " $AB = BA = I$ " (Lemma 2.54). Also, B is unique.

Proof. Consider B, B' satisfying the conditions. Then

$$B = IB = (B'A)B = B'(AB) = B'I = B'.$$

$\square$

Definition 2.57: Let A be an $m \times m$ matrix. A is invertible if and only if there exists an $m \times m$ matrix B such that $BA = I$ (or $AB = I$, or $AB = BA = I$). In this case, the matrix B is unique and called the *inverse* of A . We denote it by A^{-1} .

Observation 2.58: $(A^{-1})^{-1} = A$.

Case 1×1 .

$$A = [a] \quad \Rightarrow \quad A^{-1} = \left[\frac{1}{a}\right] \quad (\text{if } a \neq 0).$$

Case 2×2 .

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \Rightarrow \quad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (\text{if } ad - bc \neq 0).$$

Case $m \times m$. Inefficient formula (based on *Cramer's rule*); efficient computation in Chapter 3.

Lemma 2.59: Let A and B be invertible $m \times m$ matrices. Then AB is also invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Proof. Check that $B^{-1}A^{-1}$ satisfies one of the [conditions](#) for the matrix AB :

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

□

Extends to more matrices, for example $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

Lemma 2.60: Let A be an invertible $m \times m$ matrix. Then the transpose $A^{\top}$ is also invertible, and

$$(A^{\top})^{-1} = (A^{-1})^{\top}.$$